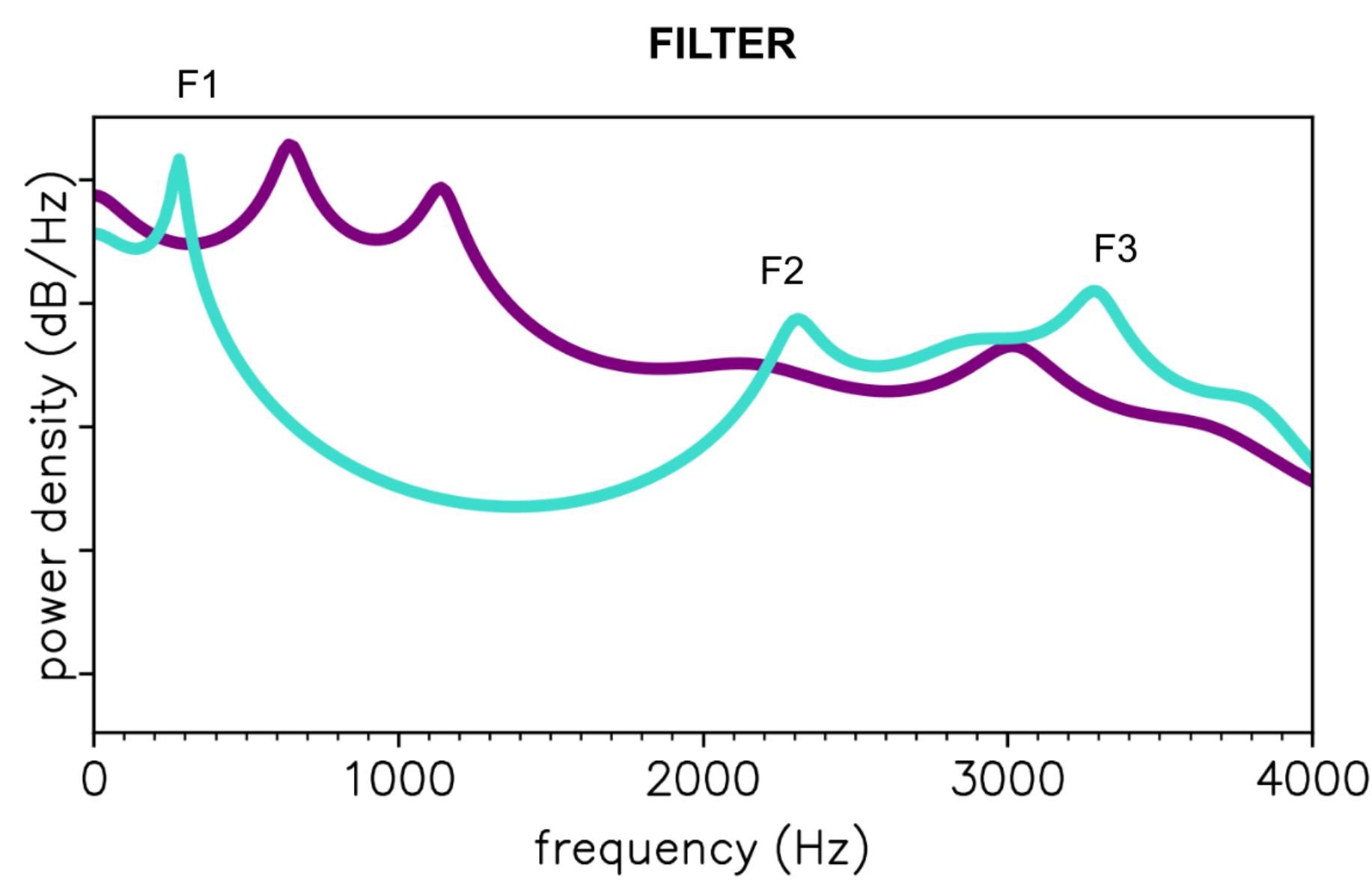
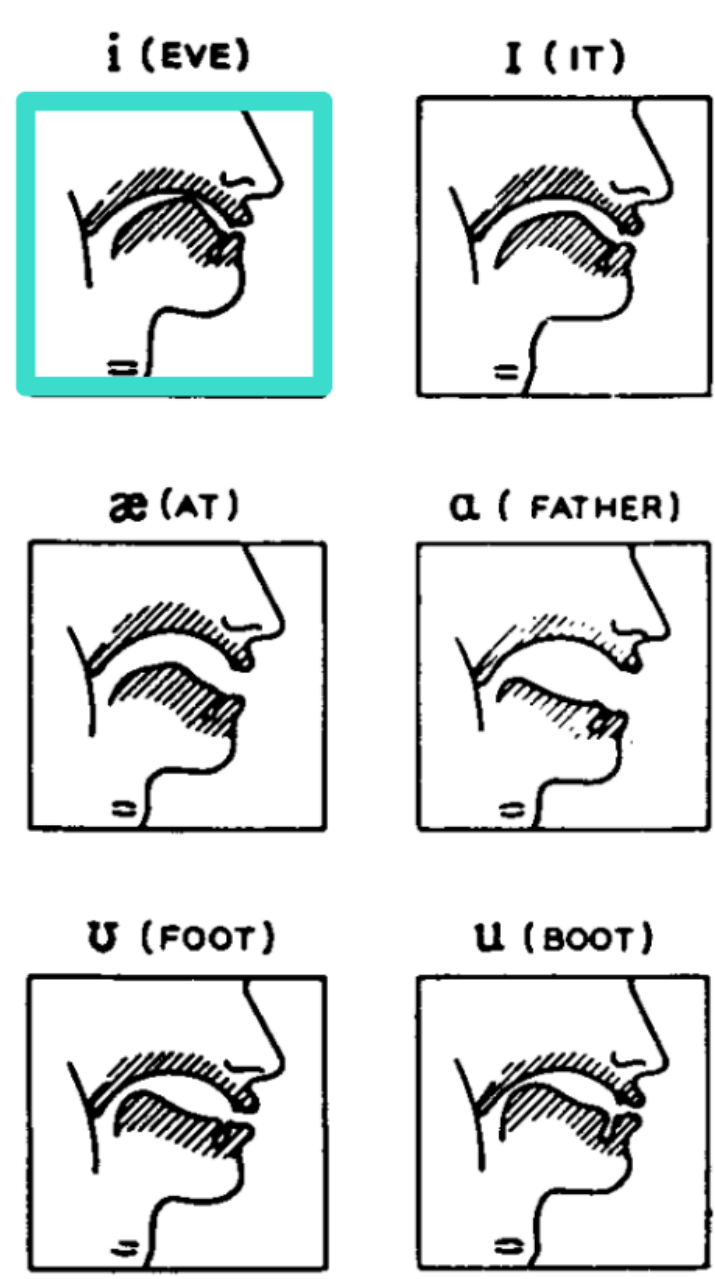


TELL YOUR FILTER WHAT YOU ALREADY KNOW

A FILTER IS A POLYNOMIAL

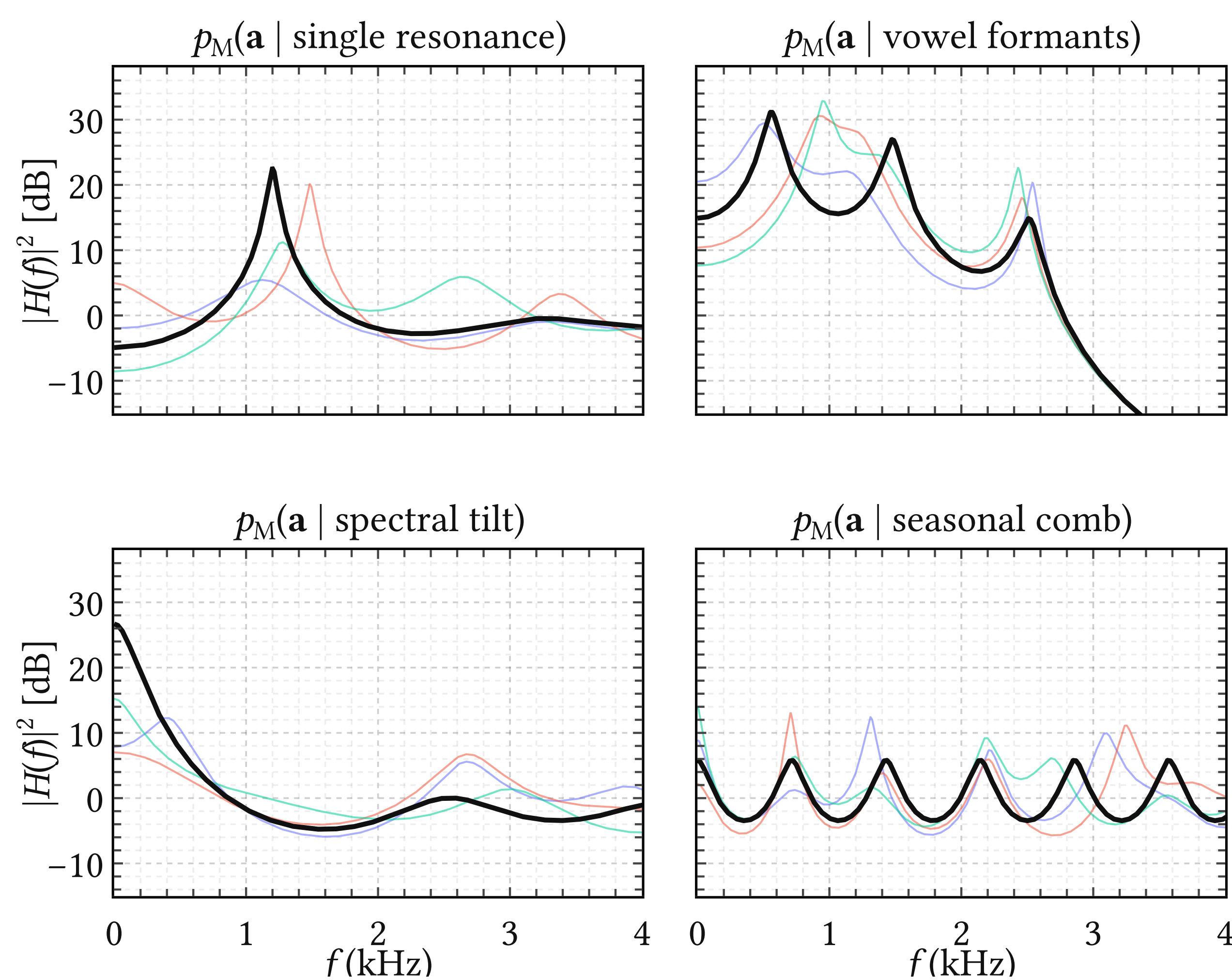
Every resonance you expect is a factor of it.

CONFIGURATION



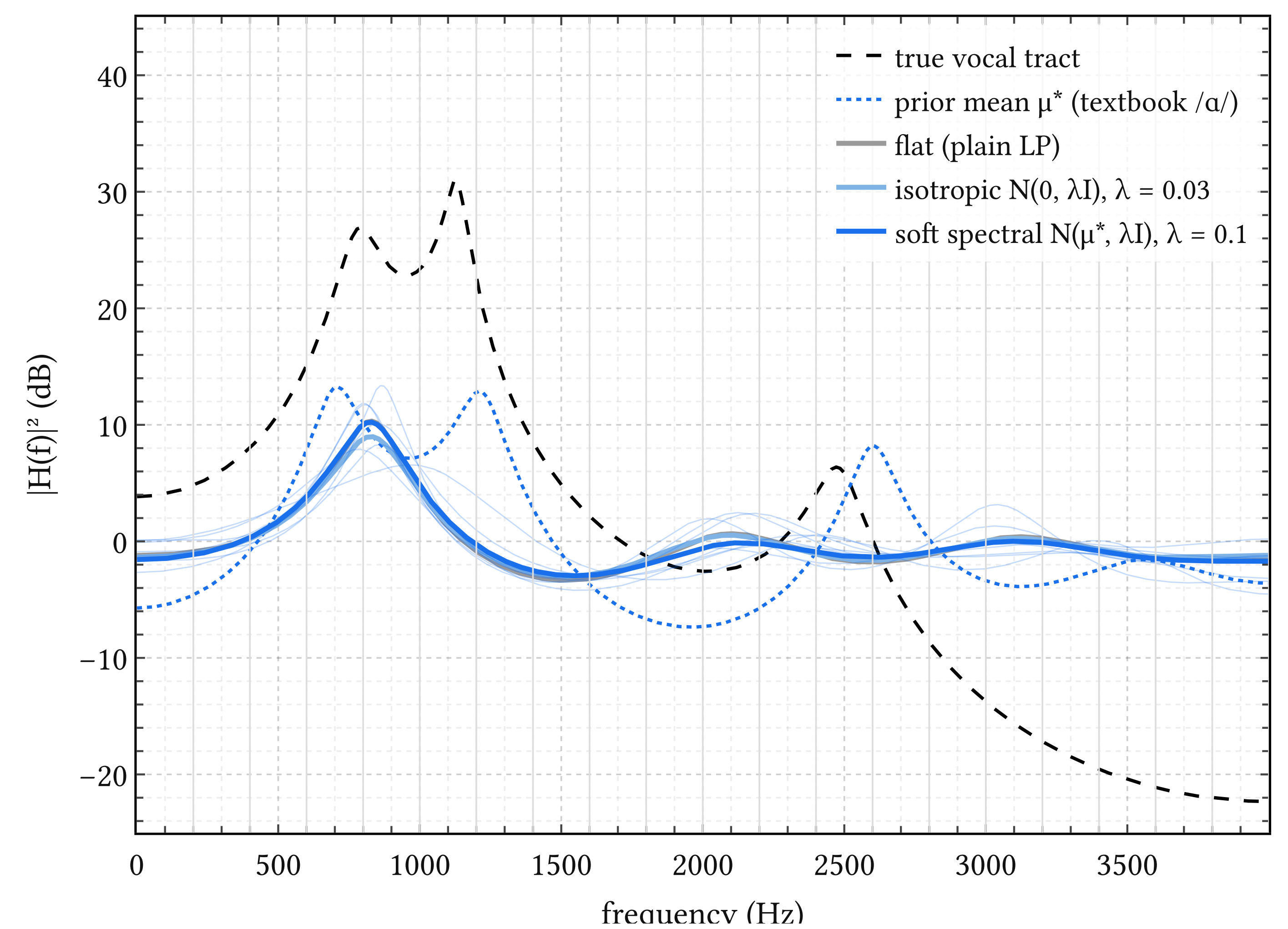
ONE UPDATE, FOUR BELIEFS

The prior before (grey) and after (black) being told about a resonance, three formants, a tilt, a harmonic comb. One closed-form update, the least change of belief (D_{KL}) that makes the polynomial divisible on average.

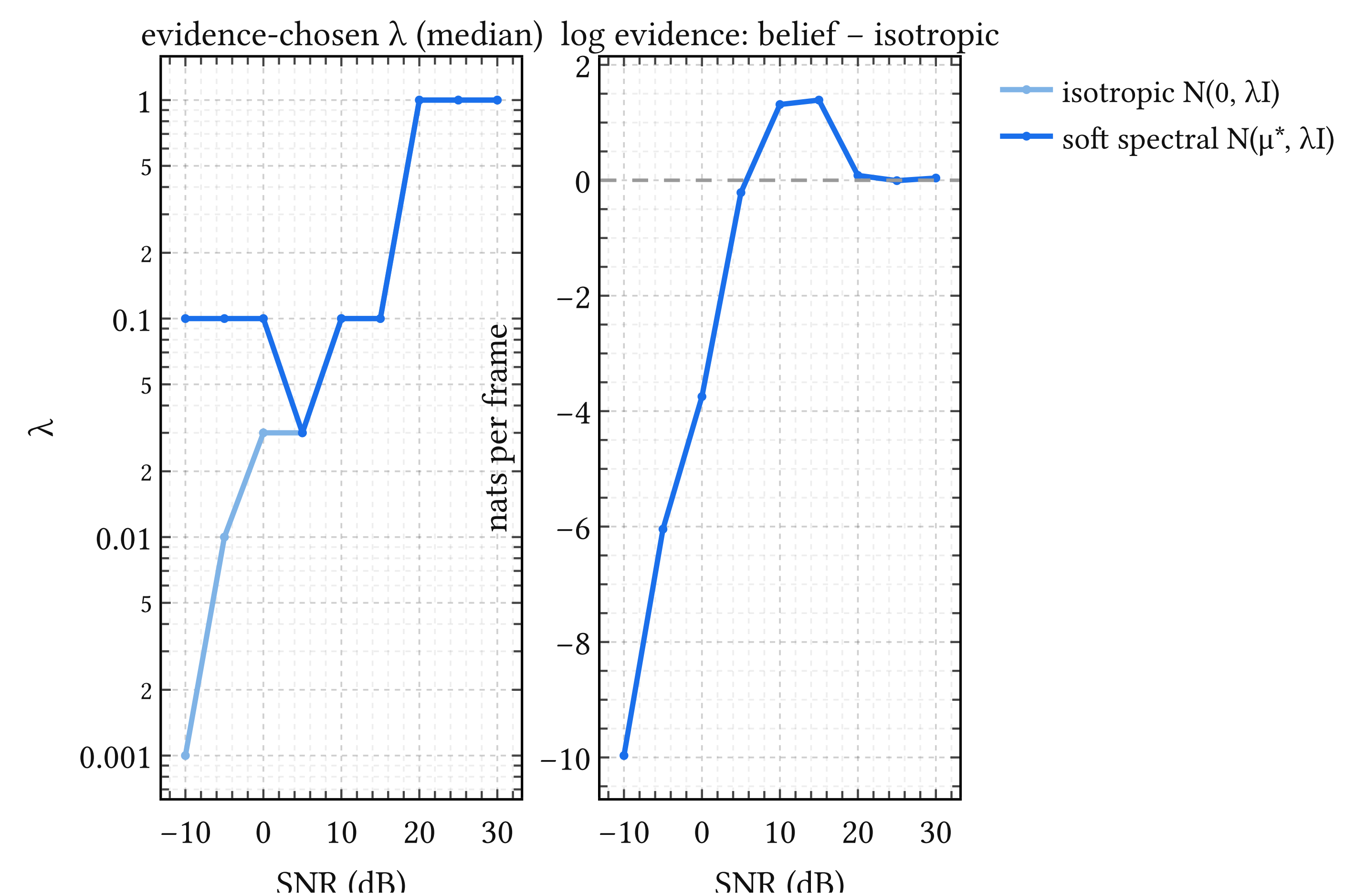


IN NOISE, IT HELPS

Synthetic /a/, 20 ms frames. The belief is the *textbook* /a/, 10% off this speaker.



One frame at 0 dB: plain linear prediction (grey) loses the first formant, the prior (blue) keeps it.



How much to trust the belief is not a knob: the evidence sets it per frame. Trusted in noise, switched off in clean frames. Below 15 dB the first-formant error drops by a fifth; above 25 dB nothing is lost.

NOT ONLY SPEECH

Bearing defect frequencies, gear-mesh harmonics, structural resonances: same polynomial, same update.

